PURDUE UNIVERSITY | Elmore Family School of Electrical and Computer Engineering

ECE 49022 – Electrical Engineering Senior Design Projects
Fall 2026 – Software Overview Report

MASHD: Multifunction Applicance for Smart Home Display

Written by: Javier Aldana

1.0 Software Overview

There are two main components to the software development of this project, the embedded firmware for the microcontroller and the iOS app development that will be controlled from an iPhone, both will interact to have an interface from the iPhone to the physical MASHD enclosure.

The embedded firmware must monitor all the sensors inside the box, measuring light, temperature, noise and movement. At the same time, it must be displaying the user-selected information on the LED matrix, which will be decided from the app. The important sensor information will be constantly available over GPIO, the built-in ADCs and any communication protocols, when the specific data is requested, an interrupt will go high and read the selected data. Furthermore, the main microprocessor will be communicating with the Wi-Fi module that is hosting its own website on a connected Wi-Fi network. In summary, the firmware listens for commands over Wi-Fi, and based on those commands puts out sensor data to the LED matrix for the viewer to use.

From the user's perspective, the web/IOS interface will allow the user to personally type out anything on the display, visualize any of the monitored data and connect to the internet to display outside weather, flights, and other interesting things to the user. The iOS app will initially connect via Bluetooth and then change to Wi-Fi after setup; the website will directly connect over Wi-Fi from the start. Once either the app or website are on the same Wi-Fi as the MASHD module, the web server hosted on the module will send out commands to the microcontroller.

2.0 Description of Algorithms

a. Digital Averaging Filter for all ADCs
   Every ADC input will have a digital filter to remove noise induced spikes. The filter works by updating a moving average over the last X inputs and treating that value like the analog signal it is measuring.
   [1] "L1: Smoothing Input," *Physical Computing*, 2025.
b. Voltage to Temperature calculation for thermistor
   The MCU will calculate the temperature from the voltage divider network and its respective temperature value from the NTC thermistor log graph. [2] NTCLE400E3472H Vishay
c. Internal Temperature to PWM strength for fan
   The fan intensity will increase based on the internal enclosure temperature by controlling pulse width matrix signal, ensuring enclosure temperature always stays below 60ºC.
d. Driving LED Matrix
   To drive 128*196-pixel array with just 13 GPIO pins as standard on LED displays, the code must use shift registers on the LED matrix to drive more than 2^13 pins.
   [3] Adafruit Industries, "64x32 RGB LED Matrix
e. System Communications
   To host a website and connect over Wi-Fi, the Wi-Fi-module will be the only device that directly interacts with the website commands, and it will then send commands to the main micro over UART.
   [4] "At Command set□," ESP
f. Ambient Light to LED Brightness
   External brightness will be monitored and the LED matrix will be dimmed as light goes down by reducing the PWM of each RGB value linearly and equally across the three.

3.0 Description of Data Structures

a. UART Communication
The Wi-Fi module should communicate with the main microprocessor over UART, an asynchronous communication protocol built in to almost every modern microcontroller. The UART communication protocol sends 10 to 11 bits made up of 1 start bit, 5 to 9 data bits, a parity bit, and a stop bit.
[5] “Analysis of universal asynchronous receiver-transmitter(uart)

b. Wireless Communication packet
To communicate between the iPhone and Wi-Fi module, the iPhone will send out HTTP requests over Wi-Fi to the module that hosts a web server. Once the data is read by the onboard module, it is converted to AT commands and is sent to the main microcontroller over UART communication.

c. Arrays
Arrays will be used in the code for moving averages in the ADC digital filters and as a framebuffers when displaying the LED arrays. The ADC array will be a 1 dimensional array [], the framebuffer is a 2d array[][] that will be used to set the next display into the shift register.
[6] P. Burgess, “Adafruit protomatter RGB matrix library,”

d. I2C Communication Protocol
The internal temperature will be monitored with an IC that uses I2C to talk to the main MCU. I2C is a two-signal serial communication protocol with a signal (SDA) and clock (SCK) line, at every rising edge, the microcontroller will read a 0, or a 1.
[1] “Ti,” A Basic Guide to I2C,

4.0 Sources Cited

[1]"L1: Smoothing Input," *Physical Computing*, 2025. https://makeabilitylab.github.io/physcomp/advancedio/smoothing-input.html (accessed Sept. 22, 2026).

[2] NTCLE400E3472H Vishay Beyschlag/draloric/BC ..., https://www.digikey.com/en/products/detail/vishay-beyschlag-draloric-bc-components/NTCLE400E3472H/2230794 (accessed Sep. 23, 2026).

[3] Adafruit Industries, "64x32 RGB LED Matrix - 6mm pitch," *adafruit*, 2020. https://www.adafruit.com/product/2276 (accessed Sept. 18, 2026).

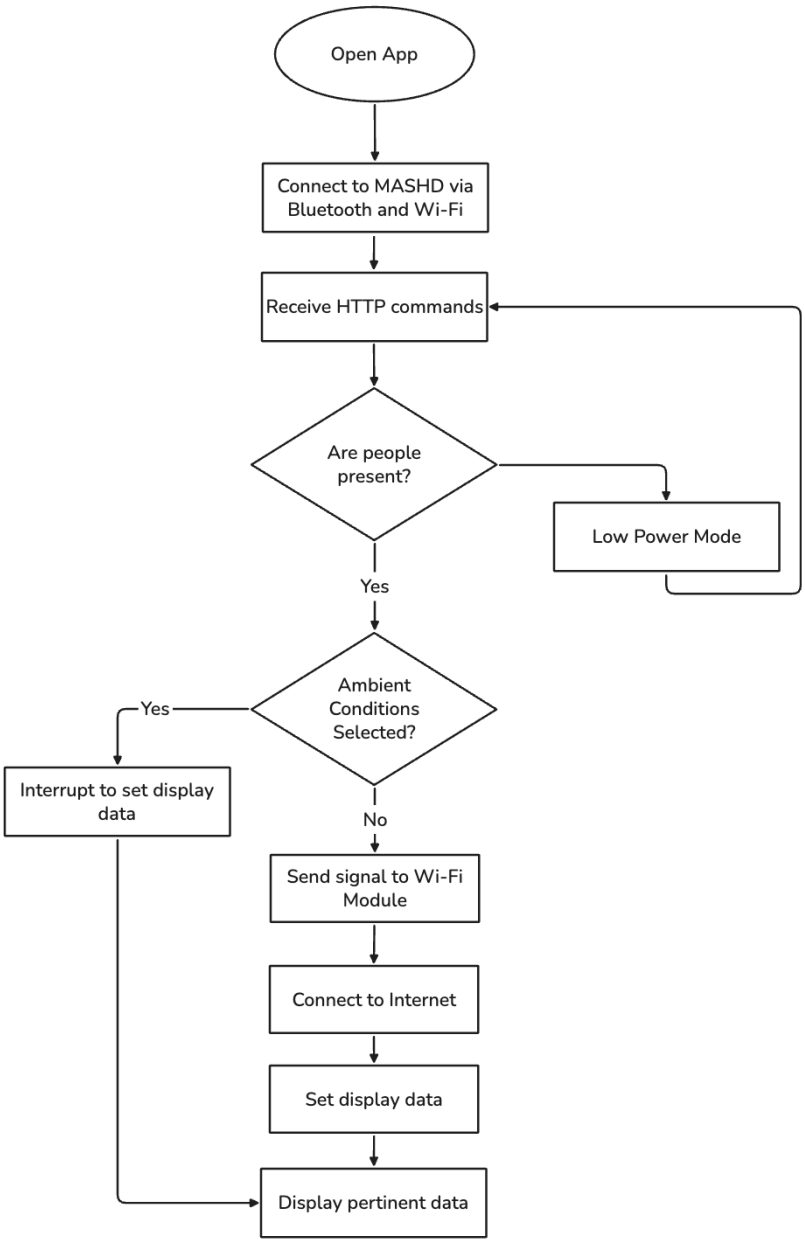
[4] "At Command set□," ESP, https://docs.espressif.com/projects/esp-at/en/latest/esp32/AT_Command_Set/index.html (accessed Sep. 23, 2026).

[5] "Analysis of universal asynchronous receiver-transmitter(uart) | IEEE conference publication | IEEE xplore," Analysis of Universal Asynchronous Receiver-Transmitter(UART), https://ieeexplore.ieee.org/document/10532820/ (accessed Sep. 24, 2026).

[6] P. Burgess, "Adafruit protomatter RGB matrix library," Adafruit Learning System, https://learn.adafruit.com/adafruit-protomatter-rgb-matrix-library/overview (accessed Sep. 25, 2026).

[7] "Ti," A Basic Guide to I2C, https://www.ti.com/lit/an/sbaa565/sbaa565.pdf?ts=1681380825945 (accessed Sep. 25, 2026).

Appendix 1: Program Flowcharts



Appendix 2: State Machine Diagrams

